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|---------------|----------------------------------------------------|
| Date          | 20/07/2026                                         |
| Team ID       | XXXXX                                              |
| Project Name  | Voice-Based Concept Understanding Analyser (VBCUA) |
| Maximum Marks | 3 Marks                                            |

Brainstorming & Idea Prioritization Template

Each team member lists as many ideas as possible without judging them at this stage.

Step 1: Brainstorm and Idea Listing

| S. No. | Team Member          | Idea / Suggestion                                | Category                    | Group No. |
|--------|----------------------|--------------------------------------------------|-----------------------------|-----------|
| 1      | Karuru Arjun         | Voice-Based Concept Understanding Analyser       | AI & Machine Learning       | 1         |
| 2      | Shaik Jubeda         | AI-Based Speech Fluency Assessment System        | Artificial Intelligence     | 1         |
| 3      | Shaik Abid Hussain   | Semantic Concept Evaluation using Speech         | Natural Language Processing | 1         |
| 4      | Aliya Bee Shaik      | Automated Spoken Concept Assessment              | Educational Technology      | 1         |
| 5      | Bavigadda Sri Ganesh | Intelligent Audio-Based Learning Feedback System | Speech Processing           | 1         |

Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

| Group No. | Final Idea                                         | Feasibility (High/Medium/Low) | Importance (High/Medium/Low) | Priority | Selected (Yes/No) |
|-----------|----------------------------------------------------|-------------------------------|------------------------------|----------|-------------------|
| 1         | Voice-Based Concept Understanding Analyser (VBCUA) | High                          | High                         | 1        | Yes               |
| 1         | AI-Based Speech Fluency Assessment System          | Medium                        | High                         | 2        | No                |
| 1         | Semantic Concept Evaluation using Speech           | Medium                        | Medium                       | 3        | No                |

Why this Idea was Selected

The Voice-Based Concept Understanding Analyser (VBCUA) was selected because it addresses the growing need for intelligent educational assessment systems that evaluate both conceptual understanding and spoken communication skills. Unlike conventional online tests that assess only textual answers, this project analyzes spoken explanations using Artificial Intelligence and Natural Language Processing techniques.

The system automatically converts speech into text, compares the explanation with predefined reference concepts using semantic similarity algorithms, evaluates speech fluency, and generates detailed feedback with performance scores. This makes the solution highly useful for educational institutions, trainers, interview preparation platforms, and e-learning environments.

The project is technically feasible using Python, Streamlit, OpenAI Whisper, Sentence-BERT, Librosa, and PDF reporting modules, making it an innovative and practical AI-based educational solution.